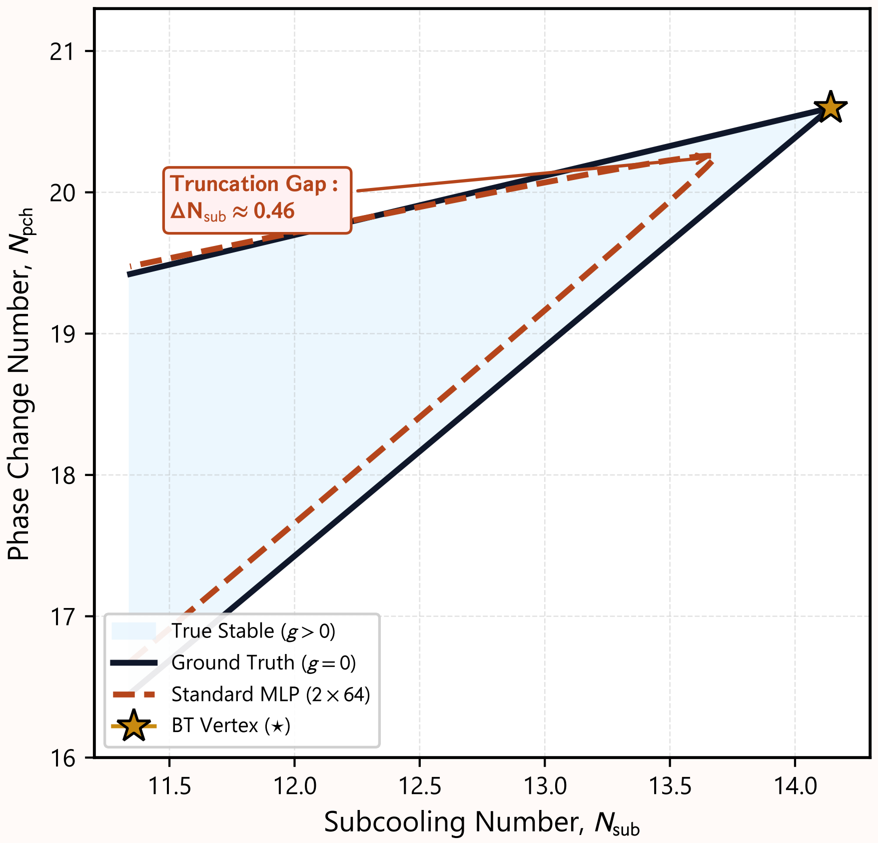
When Can a Machine-Learning Surrogate Be Trusted Near a Safety Boundary?

Codimension-2 Singularity Breakdown, Spectral Bias, and Boundary-Weighted Physics Remediation

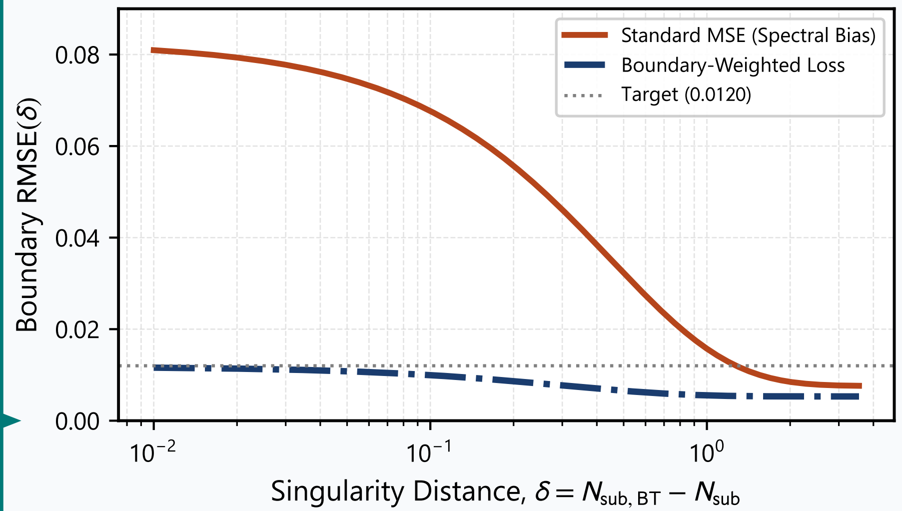
1. THE HIDDEN SAFETY PARADOX

Global Metric : $R^2 > 0.999$ (**PASSED**)
Near Cusp Metric : $R^2 < 0.000$ (**FAILED**)



Severe Cusp Truncation & False-Stable Misclassification

2. SPECTRAL BIAS & REMEDIATION

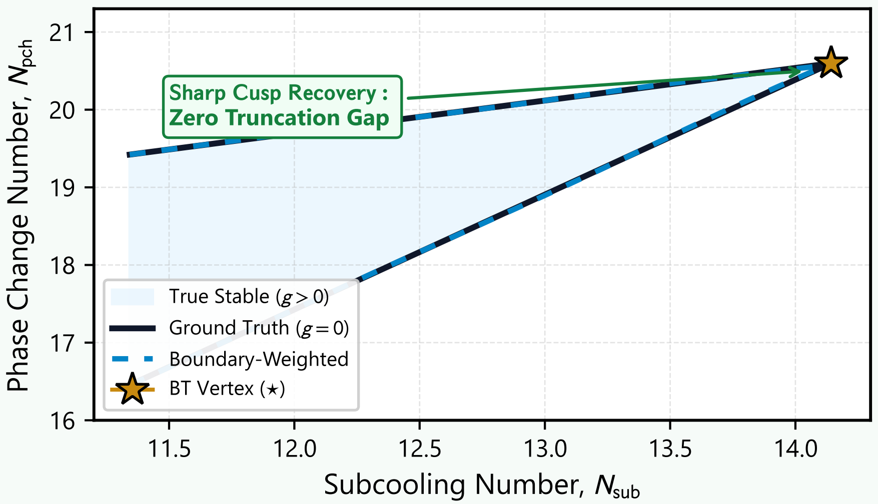


Root Cause : Spectral Bias
Standard MSE allocates capacity to smooth far-field volume,
attenuating high frequencies near singular boundaries ($\sim 11\times$ surge).

Physics – Informed Boundary Weighting :
$$\mathcal{L}_{\mathrm{BW}} = \frac{1}{|g(\mathbf{x})| + \varepsilon_w}\left(\hat{g}(\mathbf{x}) - g(\mathbf{x})\right)^2$$

Inverse-Distance Loss Restores High-Frequency Cusp Resolution

3. CERTIFIED SAFETY OUTCOME



Certified Statistical Performance :
- **48.2% Near – Boundary RMSE Reduction**
- **20/20 Random Seeds Positive** ($p = 6.68 \times 10^{-6}$)
- Global Pareto Optimum ($\mathrm{RMSE}_{\mathrm{near}} = 0.0120$)

Guaranteed Non-Parametric Reliability Certification